DATA ANALYTICS INTERNSHIP PROGRAMME 2026

Retail Sales Analytics

EXECUTIVE SUMMARY REPORT

An end to end data analytics project on the Sample Superstore dataset, covering exploratory analysis, SQL based business insights, statistical testing, time series forecasting, customer segmentation, and predictive modelling.

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Programme	ApexPlanet Software Pvt. Ltd., Data Analytics Internship 2026
Dataset	Sample Superstore Dataset (Tableau), 10,194 records, 21 features
Repository	https://github.com/KhushiVig/apexplanet-data-analytics

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Executive Summary

This report summarises a four part data analytics project built on the Sample Superstore dataset, a transactional retail dataset covering 10,194 orders and 21 fields recorded between 2023 and 2026. The project progressed from data cleaning and exploratory analysis, through SQL based business querying and interactive dashboarding, to statistical testing, time series forecasting, customer segmentation, and predictive modelling.

The core finding across every stage is consistent. Discounting is the primary driver of unprofitability, not order size or region. Sales and profit vary by product category and by customer behaviour, but not by geography or customer segment mix. The business is profitable on average, with a mean profit per order of $28.67 (95 percent confidence interval $24.16 to $33.19), shows a consistent yearly seasonal pattern, and has a highly loyal repeat customer base.

10,194 Records Analysed	**$2.33M** Total Sales	**98.5%** Customer Retention	**5** Customer Segments

Key findings at a glance

- Discounting drives losses. Discount and Profit are negatively correlated ($r = -0.22$, $p < 0.0001$), and orders above a 0.5 discount are overwhelmingly loss making.
- Technology is the strongest category, leading on both total sales ($839,893) and total profit ($146,543), while Furniture generates comparable sales ($754,748) but only $19,730 in profit.
- Regional differences are not statistically significant. ANOVA found no significant variation in mean Sales across regions ($p = 0.60$), and a t-test found no significant Profit difference between West and East ($p = 0.67$).
- Sales are strongly seasonal, with a consistent November to December peak each year and an underlying upward trend of roughly 60 percent from 2023 to 2026.
- SARIMA outperforms ARIMA for forecasting monthly sales, cutting forecast error by around 36 percent (MAPE 18.20 percent versus 28.39 percent).
- Five behavioural customer segments emerge from RFM clustering, ranging from a small, highly profitable top tier to a discount heavy segment that loses money on average.
- Churn is predictable but hard to catch early. A Decision Tree model reaches 73.9 percent accuracy but only 22 percent recall on churned customers, meaning most at risk customers currently go undetected.

Problem Statement and Objectives

The internship required an end to end analysis of retail transaction data to convert raw sales records into decisions a business could act on. The specific objectives were to clean and understand the dataset, extract business insights using SQL, build an executive dashboard, and apply statistical and machine learning methods to explain what drives profit, forecast future sales, segment customers by behaviour, and flag customers at risk of churn.

The guiding business question was straightforward. Where is the business losing money, and what should change to stop it. Every subsequent task in this report was designed to answer part of that question with evidence rather than assumption.

Data Source and Methodology

The project used the Sample Superstore dataset, sourced from Tableau's public sample data, chosen for its mix of numerical and categorical fields spanning sales, discounts, profit, shipping, and geography. The dataset required minimal cleaning. No missing values or duplicate rows were found across its 10,194 rows. Categorical columns (Ship Mode, Segment, Region, Category, Sub-Category) were converted to proper category types, and 1,913 Profit outliers and 1,183 Sales outliers were identified using the IQR method and retained, since they reflect genuine bulk orders and heavy discounts rather than data errors.

Task	Focus	Tools
1. Foundational EDA	Data cleaning, univariate and bivariate analysis, correlation review	Pandas, Seaborn, Matplotlib
2. SQL and Extraction	Business queries, window functions, joins, indexing, retention analysis	SQLite, SQLAlchemy, Pandas
3. Visualisation and Dashboarding	13 static and interactive charts, executive Power BI dashboard	Matplotlib, Seaborn, Plotly, Power BI
4. Advanced Analytics	Hypothesis testing, ARIMA/SARIMA forecasting, K-Means segmentation, regression and classification	Statsmodels, Scikit-learn, SciPy

Each stage builds on the last. The cleaned dataset from Task 1 feeds directly into the SQLite database in Task 2, the visualisations in Task 3, and the statistical models in Task 4.

Key Findings

1. Discount is the central risk factor

At zero discount, profit is overwhelmingly positive and includes every one of the dataset's highest profit orders, up to roughly $8,400. Past a 0.7 to 0.8 discount, almost every order results in a loss, including the single largest loss in the dataset, approximately -$6,600.

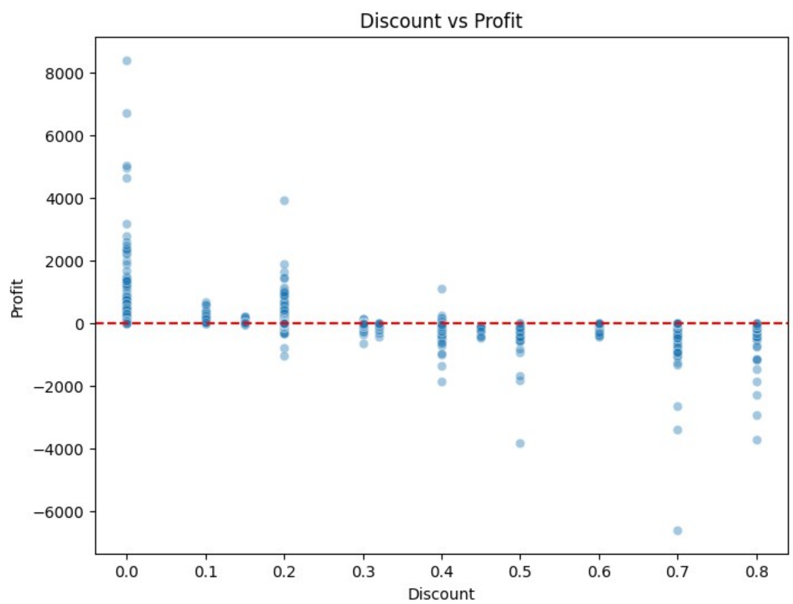

Figure 1. Discount vs Profit. Profit turns increasingly negative as discount rises past 0.3 to 0.4.

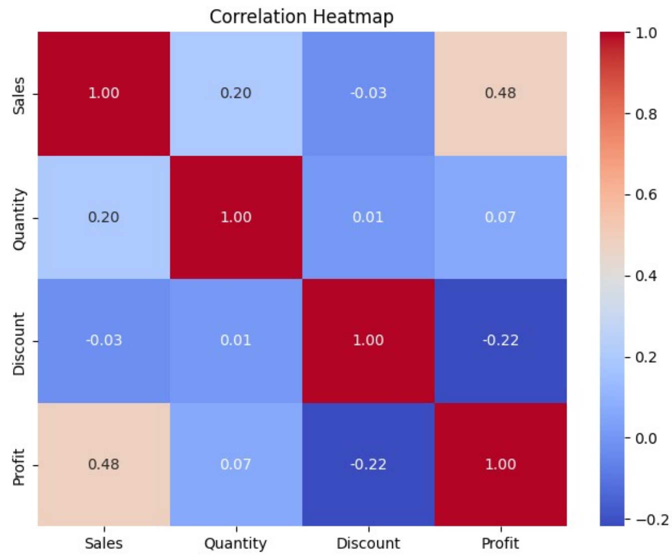

Figure 2. Correlation heatmap across Sales, Quantity, Discount, and Profit.

2. Furniture sells well but converts poorly to profit

Technology leads on both total sales ($839,893) and total profit ($146,543). Office Supplies is close behind on profit ($126,023 from $731,893 in sales). Furniture generates $754,748 in sales, comparable to the other two categories, but converts to only $19,730 in profit.

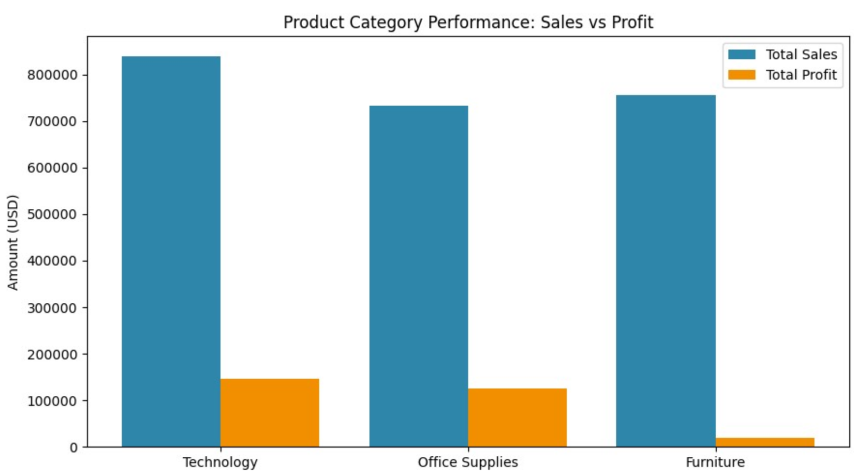

Figure 3. Sales vs Profit by category. Technology leads on both; Furniture lags badly on profit despite comparable sales.

3. Revenue peaks sharply at year end

Monthly sales show a clear year end concentration, with a sustained climb from September through November before easing in December.

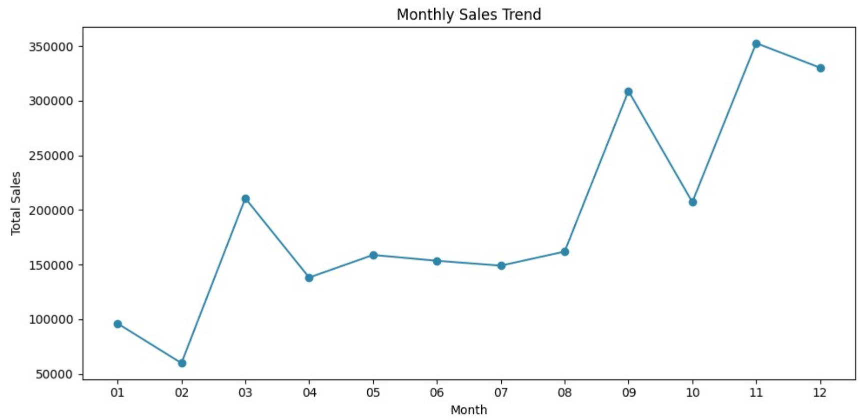

Figure 4. Monthly sales trend, showing a clear year end concentration of revenue.

4. A small group of customers drives outsized revenue

The ten highest value customers each generate between $12,000 and $25,000 in lifetime revenue, led by Sean Miller at $25,043.05. More broadly, customer retention is exceptionally strong. Of 800 unique customers, 788 (98.5 percent) placed more than one order.

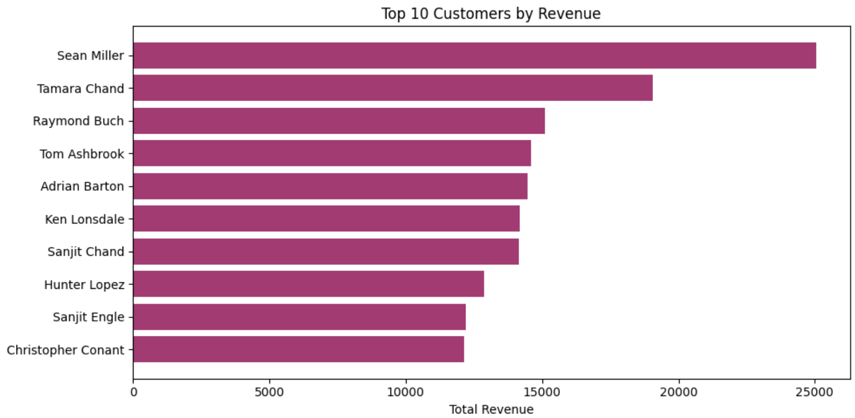

Figure 5. Top 10 customers by total revenue, led by Sean Miller at $25,043.05.

Statistical Analysis Results

Hypothesis testing

Three hypothesis tests were run to check whether commonly assumed regional or segment effects are statistically real.

- T-test, West vs East profit: $t = 0.4196$, $p = 0.6748$. No significant difference.
- Chi-square, Segment vs Region: chi-square = 6.1988, dof = 6, $p = 0.4013$. Segment mix is independent of region.
- ANOVA, Sales across regions: $F = 0.6204$, $p = 0.6017$. No significant difference in mean order value by region.

Together these results indicate that regional variation in total revenue is driven by order volume, not by any real difference in average order value or profitability by geography.

Time series decomposition and forecasting

Decomposing the monthly sales series confirms a steady upward trend, from roughly $40,000 per month in 2023 to over $65,000 per month by late 2026, with a consistent yearly seasonal shape. The original monthly series was already stationary by the ADF test (statistic -4.4375, $p = 0.0003$), and first differencing produced an even more strongly stationary series (statistic -8.9944, $p < 0.0001$), supporting the use of $d = 1$ in the ARIMA specification.

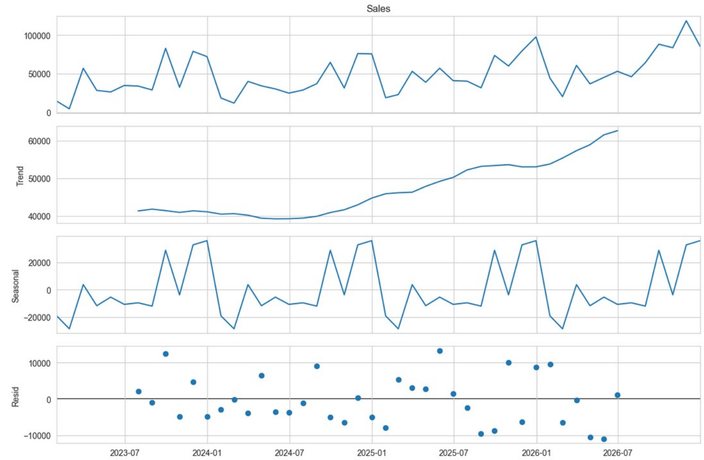

Figure 6. Seasonal decomposition of monthly sales: trend, yearly seasonality, and residuals.

Because plain ARIMA cannot represent the yearly seasonal cycle, it forecasts a near flat line and produces the larger error (MAE $25,158.73, RMSE $33,358.76, MAPE 28.39 percent). SARIMA, which explicitly models the 12 month seasonal component, cuts the error substantially (MAE $14,183.07, RMSE $17,192.27, MAPE 18.20 percent).

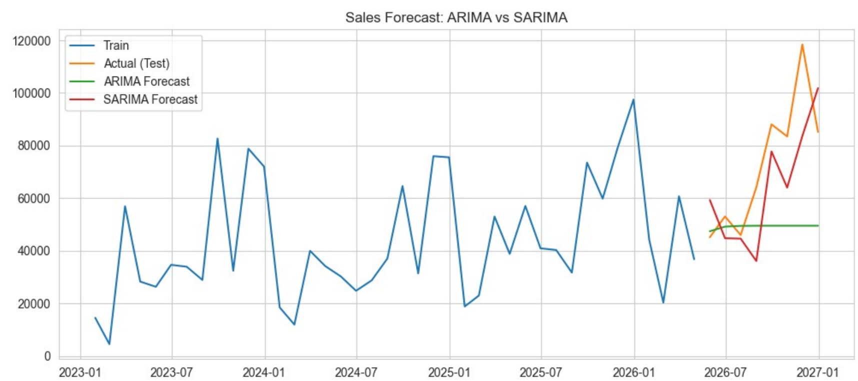

Figure 7. ARIMA vs SARIMA forecasts against actual held out monthly sales.

Customer segmentation (RFM and K-Means)

Each of the 800 customers was profiled on five behavioural features, Recency, Frequency, Monetary value, Average Discount, and Total Profit, then clustered using K-Means. Silhouette analysis favoured k = 3 (score approximately 0.275), but k = 5 was selected to produce more actionable, business relevant segments.

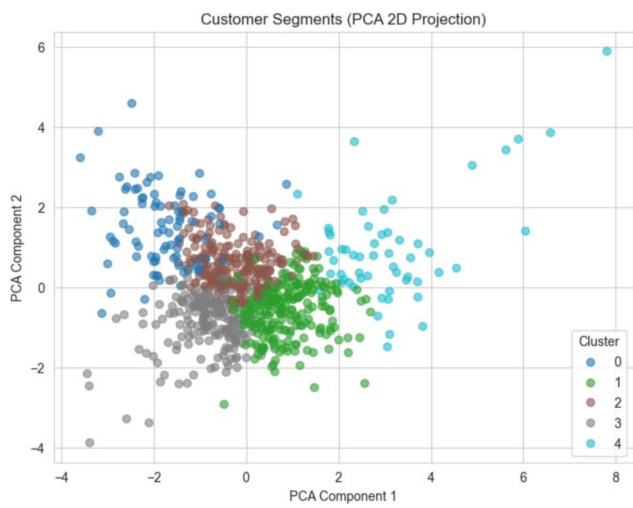

Figure 8. Customer segments in 2D PCA space. The two components explain 38.9 percent and 24.4 percent of variance.

Segment	Size	Share	Total Profit / Customer
Cluster 4, Top Tier	55	6.9%	$2,329
Cluster 1, Core Active	247	30.9%	Positive, moderate
Cluster 2, Efficient Buyers	226	28.3%	Positive, low discount
Cluster 3, Discount Heavy	192	24.0%	-$182.60
Cluster 0, Lapsed	84	10.5%	Low, inactive

Cluster 4 buys at close to full price and drives the highest profit per customer by a wide margin. Cluster 3 is the only segment with negative average profit, driven by the highest average discount rate at 0.25, directly consistent with the discount to profit relationship identified earlier in this report.

Predictive modelling

A linear regression model predicting Sales from Quantity, Discount, Profit, and one-hot encoded Segment, Region, Category, and Ship Mode achieved R-squared = 0.5266 (MAE $218.76, RMSE $497.12), explaining roughly half the variance in Sales. For churn, customers inactive for more than 180 days were labelled churned as a recency based proxy, yielding a 74.6 percent active and 25.4 percent churned split.

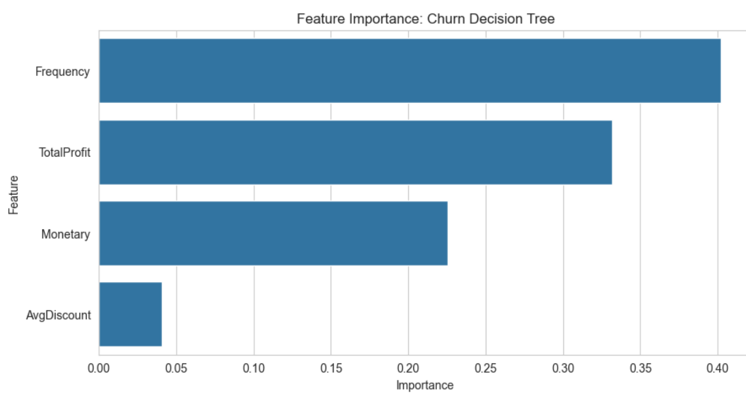

Figure 9. Feature importance for the churn Decision Tree. Frequency and Total Profit dominate.

73.9%	47.4%	22.0%	0.67
Decision Tree Accuracy	Precision	Recall	ROC-AUC

Both Logistic Regression (72.67 percent accuracy) and the Decision Tree (73.91 percent accuracy) sit close to the 74.6 percent always-predict-active baseline, and both show low recall, meaning most true churners currently go undetected. Frequency (0.40) and Total Profit (0.33) are the strongest predictors of churn, while discount habits contribute little (0.04).

Dashboard Overview

An interactive Power BI executive dashboard was built on the cleaned dataset, comprising four KPI cards (revenue, orders, customers, profit), a monthly sales trend line, a category breakdown donut chart, a sales by region bar chart, and a top 10 products by sales bar chart.

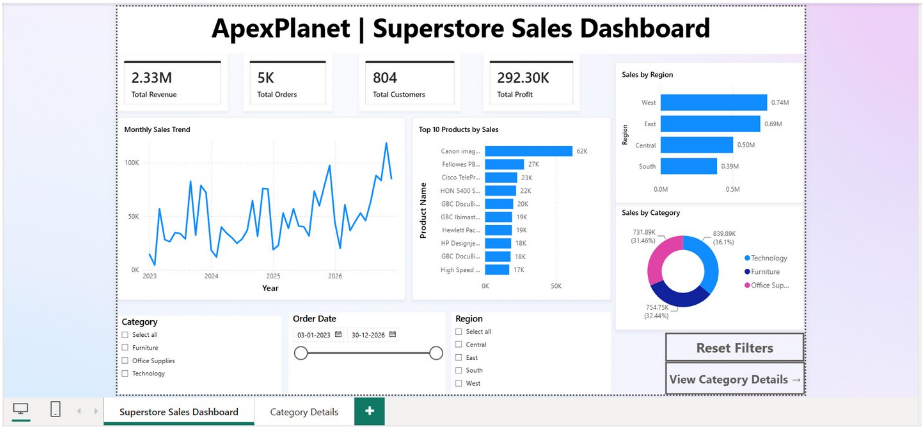

Figure 10. Executive dashboard overview, Power BI.

A drill-through Category Details page provides product level detail, accessed by right-clicking any category slice, alongside page navigation buttons and a Reset Filters bookmark.

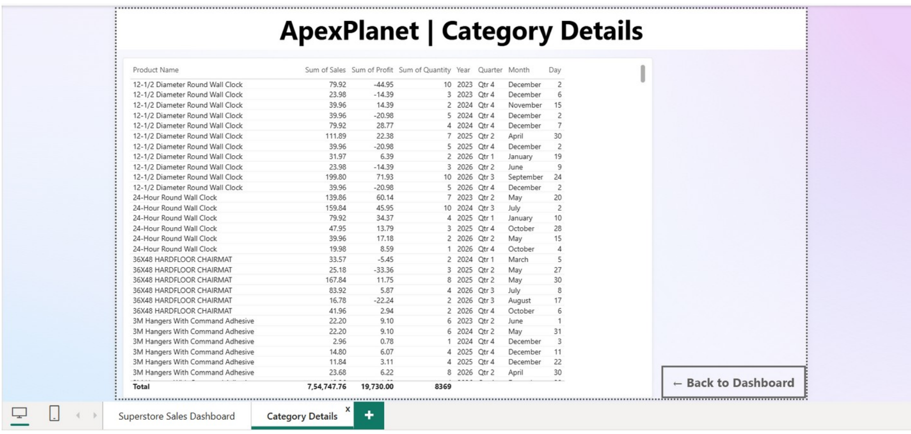

Figure 11. Category Details drill-through page.

Thirteen additional static and interactive visualisations were produced in Python across Matplotlib, Seaborn, and Plotly. The dashboard file and all supporting notebooks are available in the project GitHub repository.

Business Recommendations

- Tighten discount governance above the 0.5 threshold. Discounting is the clearest and most consistent driver of unprofitability across the EDA, SQL analysis, and regression. A discount approval tier or cap for orders in the 0.5 to 0.8 range would directly protect margin.
- Investigate Furniture's profit efficiency specifically. It matches Technology and Office Supplies on sales volume but returns barely a tenth of their profit, pointing to a pricing, cost, or discounting issue unique to that category rather than a demand problem.
- Prioritise Cluster 4 and Cluster 3 customers for opposite reasons. Cluster 4, with 55 customers and outsized profit, merits a loyalty programme. Cluster 3, with 192 customers and the only segment losing money, merits an immediate discount policy review.
- Plan inventory and staffing around the confirmed November to December peak rather than treating all months equally, and adopt SARIMA over ARIMA for operational sales forecasting given its materially lower error.
- Do not localise pricing or marketing strategy by region. Hypothesis testing found no statistically significant regional effect on profit or average order value, so regional revenue gaps are a volume story, not a pricing or segment mix story.

Limitations and Future Scope

Limitations

- The dataset is a static, single source sample and may not reflect real world seasonality, competition, or macroeconomic effects.
- Outliers in Sales (11.6 percent) and Profit (18.8 percent) were retained as genuine business events, so both the regression and clustering models must absorb a wide value range.
- The churn label is a recency based proxy, not a confirmed cancellation event, so some flagged customers may simply be infrequent shoppers rather than lost customers.
- Both churn models show low recall (12 to 22 percent). They are useful for understanding churn drivers but are not yet reliable enough to automatically trigger retention interventions.
- The Sales regression explains about 53 percent of variance and underperforms specifically on high value orders, which are exactly the orders most worth predicting accurately.

Future scope

- Incorporate trend based and time aware features, not just static RFM, into the churn model to improve recall.
- Extend the SARIMA forecast with more historical cycles as new data becomes available, and benchmark against alternative forecasting methods such as Prophet or gradient boosted models.
- Automate the analytics pipeline, covering extraction, cleaning, and KPI refresh, to keep the dashboard and models current without manual reruns.
- Apply causal inference methods to the Discount-Profit relationship to move from correlation to a defensible causal estimate of discounting's true margin impact.

Prepared by Khushi Vig, ApexPlanet Software Pvt. Ltd. Data Analytics Internship Programme 2026.

Project repository: https://github.com/KhushiVig/apexplanet-data-analytics